

2(a)(i)	<i>any 2 from:</i> 1 <i>idea of</i> randomising the planting of the different types of cotton plants ; 2 <i>idea of</i> large number of samples examined ; 3 AVP ;	2
2(a)(ii)	<i>any 2 from:</i> 1 (ambient) temperature ; 2 (sun)light (availability) ; 3 rainfall / soil water ; A availability of water 4 humidity ; 5 wind ; 6 (variation in) soil pH ; 7 other soil feature e.g. drainage / fertility / nutrients already present / aeration / type of soil ; 8 pests other than that, bollworm / disease ; 9 AVP ;	2
2(b)(i)	<i>any 1 from:</i> 1 to show the closeness of the mean to the, true / population, mean ; 2 for drawing error bars on graph(s) ;	1

Question	Answer	Marks
2(b)(ii)	<i>any 2 from:</i> 1 compares the means of two sets of data (for each year) ; 2 the data is continuous ; 3 data is normally distributed ;	2
2(b)(iii)	<i>any 3 from:</i> 1 egg density in Bt cotton and non-Bt cotton decreases (from approx 800 per plant to approx 200) ; 2 egg density increases from 2004 ; 3 larvae density on Bt cotton, fluctuates / shows little change ; 4 larvae density on non-Bt cotton decreases ; 5 AVP ;	3
2(c)(i)	<i>any 1 from:</i> 1 time when sampling is carried out / <i>idea that</i> sampling occurs during breeding season / AW ; 2 area of each field sampled ; 3 number of fields sampled ; 4 AVP ; e.g. training / experience, of those doing the sampling	1
2(c)(ii)	<i>idea that</i> the use of Bt cotton kills larvae so (each year) fewer survive to become adults and lay eggs on crops other than cotton / AW ;	1